

# Embedding-Aware Cooperative LiDAR Perception

## Spatial TrackFormer Association and SWFormer Detector Exploration

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TrackFormer [1] is adapted from temporal tracking to same-timestamp cross-agent association in OPV2V.

# What We Built

## Detector and data

- ▶ OpenCOOD PointPillars runs on the target hardware.
- ▶ Clean late fusion reproduces  $AP@0.7 \approx 0.78$ .
- ▶ We cache detections, identities, confidence, box geometry, and every vehicle-covered BEV feature cell.

## Association objective

- ▶ Given a source-agent detection and an ego-agent detection at the same timestamp, decide whether they are the same physical vehicle.
- ▶ Transform box geometry into the ego frame before comparison.
- ▶ Output a normalized 128-D association embedding per proposal.

**Key correction:** the original MLP read only one BEV feature cell; the final model reads all cells covered by the vehicle box.

# SWFormer: Why Use a Sparse-Window Transformer?

## The LiDAR problem

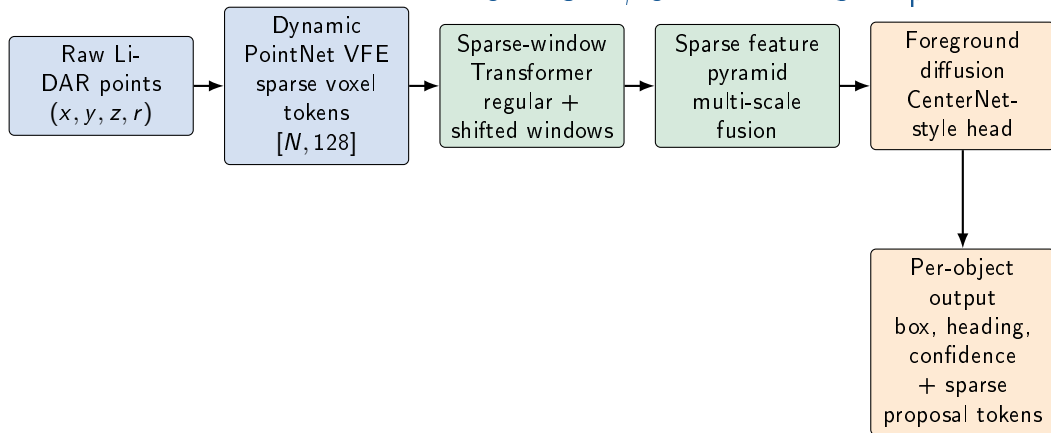
- ▶ A driving scene spans a large ground-plane area, but only a small fraction of cells contain LiDAR points.
- ▶ A dense BEV CNN still computes over many empty cells; global attention over all cells would be prohibitively expensive.

## SWFormer contribution [2]

- ▶ Keep only occupied voxels as sparse tokens.
- ▶ Apply self-attention inside small spatial windows, then shift windows so neighbouring regions can exchange information.
- ▶ Fuse multiple sparse resolutions and diffuse foreground information before predicting object centers and boxes.

**High-level goal:** obtain Transformer context without paying for dense computation over empty LiDAR space.

# SWFormer Detector Architecture: One CAV, One LiDAR Sweep



- ▶ A **window** is a local  $10 \times 10$  BEV neighbourhood; attention compares occupied tokens only within that window. Shifted windows connect neighbouring windows on the next block.
- ▶ The feature pyramid provides both fine spatial detail and broader scene context. Voxel diffusion supplies nearby context where LiDAR points are sparse around an object.
- ▶ Our current OpenCOOD experiment uses one sweep per CAV; it does not yet exploit SWFormer's optional multi-sweep temporal input.

# How SWFormer Learns 3D Detection

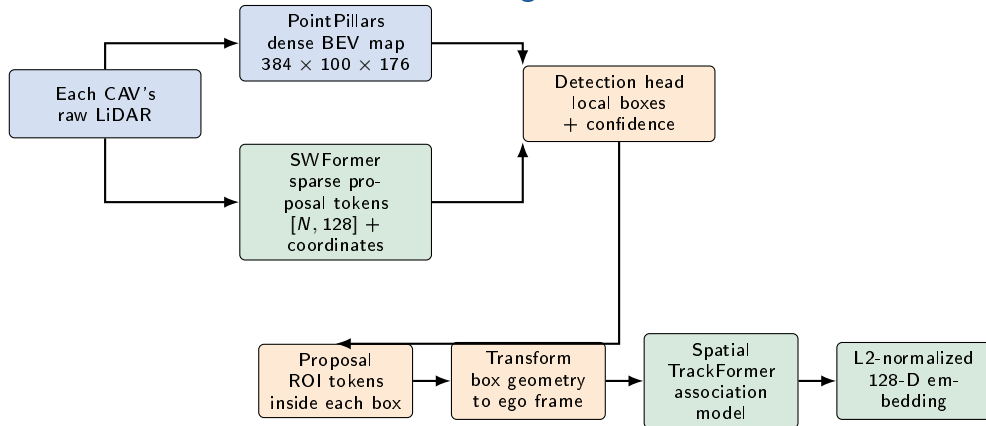
For each sparse BEV token, SWFormer predicts foreground, an object-center heatmap, box parameters, and heading. The implemented objective is

$$\mathcal{L}_{\text{SW}} = 200 \mathcal{L}_{\text{seg}}^{\text{focal}} + 10 \mathcal{L}_{\text{center}}^{\text{focal}} + \mathcal{L}_{\text{box}}^{\text{Smooth-L1}} + \mathcal{L}_{\text{heading}}^{\text{CE+Smooth-L1}} + \mathcal{L}_{\text{IoU}}^{\text{3D}}.$$

- ▶  $\mathcal{L}_{\text{seg}}$ : classify sparse cells as foreground or background; focal weighting handles the large background majority.
- ▶  $\mathcal{L}_{\text{center}}$ : make each vehicle center a confident local heatmap peak.
- ▶  $\mathcal{L}_{\text{box}}$ : regress center offsets and  $h, w, l$ .
- ▶  $\mathcal{L}_{\text{heading}}$ : classify one of 12 yaw bins and regress the residual within that bin.
- ▶  $\mathcal{L}_{\text{IoU}}^{\text{3D}}$ : directly rewards overlap between the decoded oriented prediction and its ground-truth box.
- ▶ The final confidence score is used to rank detections for the precision–recall curve and AP.

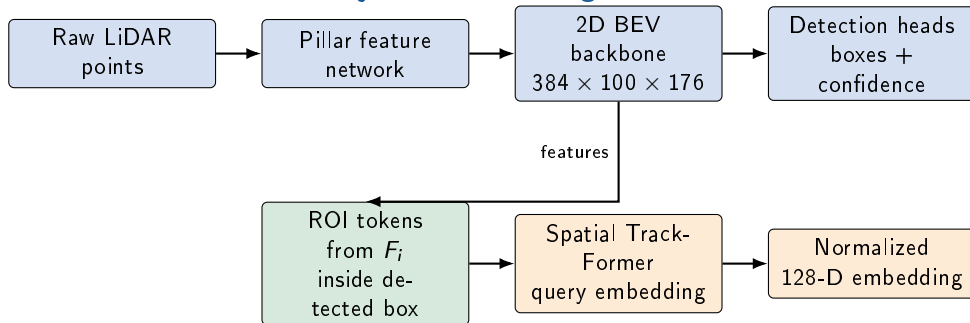
**Important:** lower training loss is useful, but validation AP measures the actual detection-and-ranking quality we report.

## Detector Features → Shared Embedding Interface



- ▶ **Shared contract:** whichever detector is used, it supplies a detected box, confidence, and the proposal-aligned feature tokens inside that object region.
- ▶ The source and ego boxes are represented in the same ego coordinate frame before association. Spatial TrackFormer then maps object evidence to a 128-D vector; similar vectors support cross-agent grouping.
- ▶ PointPillars currently supplies the trained embedding cache. SWFormer now exposes its native sparse proposal tokens, enabling the same downstream association interface in a future experiment.

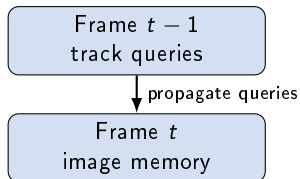
# From LiDAR Points to an Object Embedding



- ▶ **Pillar feature network:** a point-wise MLP and max pooling convert all points in one vertical XY cell into one learned pillar descriptor.
- ▶ **2D BEV backbone:** a CNN scatters these descriptors onto the ground plane and combines neighbouring pillars into contextual features.
- ▶ 384 is the number of learned channels, *not* the number or size of pillars;  $100 \times 176$  is the final spatial grid after the backbone's strides, starting from the configured LiDAR range and 0.4 m voxel grid.
- ▶ **Detection heads** are small convolutional layers that predict a vehicle score and box offsets at every BEV location/anchor. **ROI** means *Region of Interest*: the 6–21 BEV cells covered by that box.

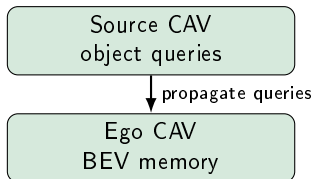
# TrackFormer Training: Temporal Tracking → Spatial Association

## Original TrackFormer



Frame  $t - 1$  detections spawn queries that carry identity into frame  $t$ .

## Our spatial adaptation



Source-CAV detections spawn queries that attend to ego-CAV BEV memory at the same timestamp.

**Training objective from the TrackFormer paper.** New objects use Hungarian matching; persistent tracks are matched by their known identity:

$$\hat{\sigma} = \arg \min_{\sigma} \sum_i \left[ -\lambda_{\text{cls}} \hat{p}_{\sigma(i)}(c_i) + C_{\text{box}}(b_i, \hat{b}_{\sigma(i)}) \right], \quad C_{\text{box}} = \lambda_1 \|b_i - \hat{b}_{\sigma(i)}\|_1 + \lambda_{\text{giou}} C_{\text{giou}}.$$

$$\mathcal{L}_{\text{MOT}} = \sum_j \left[ \text{CE}(c_j, \hat{p}_j) + \mathbb{I}_{j \in \pi} (\lambda_1 \mathcal{L}_1 + \lambda_{\text{giou}} \mathcal{L}_{\text{giou}}) \right], \quad c_j = 0 \text{ for unmatched queries.}$$

**Why this can work:** source queries attend to ego BEV memory; shared-frame geometry disambiguates similar vehicles. False negative and false positive queries teach recovery and rejection.

# Exact Cross-Agent TrackFormer Pipeline

For source agent  $i$ :  $F_i \in \mathbb{R}^{384 \times 100 \times 176}$  is its local BEV map and  $P_i = \{(b_{i,k}^{\text{local}}, s_{i,k})\}$  are its local detected boxes and confidence scores.

## Definitions: one proposal $k$ from source agent $i$

- ▶  $\mathcal{R}_{i,k}$ : **index set** of BEV cells covered by the rotated detected box  $b_{i,k}$ ; it is not a feature vector.
- ▶  $X_{i,k} = [F_i(r_1); \dots; F_i(r_m)] \in \mathbb{R}^{m \times 384}$ : the actual **ROI token data** selected from  $F_i$  using  $\mathcal{R}_{i,k}$ , with  $m \approx 6\text{--}21$  cells.
- ▶  $g_{i,k} \in \mathbb{R}^9$ : the normalized ego-frame geometry and confidence  $[x_e, y_e, z_e, h, w, l, \sin \psi, \cos \psi, s_{i,k}]$ .
- ▶  $h_{i,k} \in \mathbb{R}^{256}$ : learned final decoder **query state**; an internal float vector, not a box or raw LiDAR data.
- ▶  $z_{i,k} \in \mathbb{R}^{128}$ : the L2-normalized **association embedding** computed from  $h_{i,k}$  and  $g_{i,k}$ .

## Process and communication

**Source:**  $F_i, P_i \rightarrow \mathcal{R}_{i,k}, X_{i,k}, g_{i,k} \rightarrow \text{TrackFormer} \rightarrow (h_{i,k}, z_{i,k})$ ; send selected query states, ego-frame boxes, scores, and embeddings—not the full  $F_i$  map.

**Ego:** builds  $X_{e,k}, g_{e,k}$ , propagates received  $h_{i,k}$  through its BEV memory, then associates source and ego proposals. The detection head defines each ROI and explicitly conditions TrackFormer through  $g_{i,k}$ .

## Embedding Results and Current Conclusion

| Model / evaluation                                    | Top-1 association accuracy |
|-------------------------------------------------------|----------------------------|
| Single-cell MLP, held-out cache                       | $\approx 0.21$             |
| Spatial TrackFormer + geometry head, clean validation | 0.915                      |
| Spatial TrackFormer + geometry head, clean OPV2V test | 0.891                      |
| Same clean model, noisy validation                    | 0.907                      |
| Same clean model, noisy OPV2V test                    | 0.891                      |

### Takeaway

Using all vehicle-covered BEV cells and spatial TrackFormer query propagation substantially improves cross-agent identity association. The embedding model is strong in isolation; the remaining work is to translate this association quality into better end-to-end fusion recall under noisy geometry.



T. Meinhardt et al., “TrackFormer: Multi-Object Tracking with Transformers,” CVPR, 2022.



P. Sun et al., “SWFormer: Sparse Window Transformer for 3D Object Detection in Point Clouds,” ECCV, 2022.